

to minimise the problem of trailing wires when plugging the device in to recharge the battery.

PC kit for kids

At a New York toy fair, PeeWee PC unveiled a kit that transforms any computer into an age-appropriate PC for children. The kit boasts of a USB drive, a PeeWee security suite for online and offline safety, and various fun games and educational software. Tweaked to conform to low-cost netbooks, the PeeWee kit is surely a kid's best friend. It can be customised for three levels of students: Pre-K, early elementary or upper elementary.

The security suite includes 'PeeWee Patrol' for computer access controls and 'PeeWee Privacy, ostensibly meant for Internet privacy. The software lets parents keep a watchful eye on their children's use of the laptop and the content they access. It also helps block inappropriate sites with any version of any Web browser, shows browsing history and even lets parents control the laptop from a remote location.

Water-resistant LEDs

Avago Technologies claims to have developed the industry's first high-brightness, water-resistant, tricolour surface-mount light-emitting diodes (LEDs) for indoor and outdoor full-colour signs and displays. These compact LEDs are ideal for use in electronic video screens and advertising signs used in transit malls, airports and stadium scoreboards.

The ASMT-YTB0 LEDs provide full-colour display application designers with a 115-degree viewing angle and better screen resolution (greater than 12mm pixel size).

The compact ASMT-QTC0 package incorporates a separate heat path for each LED dice to enable it to be driven at higher currents. As a result, these SMT LEDs can operate in a wide range of environmental conditions to provide customers with high reliability.

Both tricolour LEDs are compatible with reflow-soldering processes and have a moisture-sensitivity level of



100mm diameter graphene wafer

'2a,' which makes them ideal for use in SMT production environments.

100mm graphene wafers

Using silicon sublimation, two researchers at the Electro-Optics Centre (EOC) of Pennsylvania State University have produced 100mm graphene wafers.

To achieve this, David Snyder and Randy Cavalero heated silicon carbide wafers in a physical vapour transport furnace until the silicon migrated away from the surface, leaving behind a layer of carbon that formed into a one-to-two-atom-thick film of graphene on the wafer surface.

According to EOC materials scientist Joshua Robinson, RF FETs have been made on the wafers. However, the electron mobility of the EOC wafers has not been established so far.

Next-generation GPU

Nvidia has announced high-performance parallel computing GPU architecture. The Tesla C2070 and C2050 are PCI Express x16 cards that can turn a Windows or Linux workstation to perform like a small cluster with up

to 640 Gigafllops of performance. The devices do not run Windows or Linux directly. Rather, these are designed to run C, C++, OpenCL, DirectCompute or Fortran.

The C2070 and C2050 are PCI Express Gen2 cards and include either 3 GB or 6 GB of onboard GDDR5 memory, respectively. The GPU consumes 190 watts.

MCUs for cars

Renesas Technology's SH7450 Series MCUs for advanced driver assistance applications combine on-chip flash memory and high-speed operation. The specifications of the SH74552 and SH74562 have been carefully selected for systems intended for low-priced vehicles such as eco-friendly cars



Nvidia Tesla C2070 GPU